A second-order resonance scale for coboundary kicks of a fixed cat map

WoW research programme

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Status, precedent, and statement

This note gives a project proof, with separately scoped finite checks. It is not a claim of world priority or a complete Lean formalization. Residual perturbations, removal of a coboundary at first order, and a second-order commutator as the effective perturbation are established mechanisms in the quantum-fidelity literature: Prosen and Znidaric, *Phys. Rev. Lett.* **94** (2005), 044101, especially equations (4)–(7) and (12) of the primary preprint quant-ph/0401142. The specific assertions proved here are a uniform finite operator estimate, a two-species arithmetic collision classification, and deterministic spectral limits along the Pell recurrence sequence. Fidelity decay in a mixing approximation must not be substituted for these spectral assertions.

Set

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad \alpha = 1 + \sqrt{2}, \quad \lambda = 3 + 2\sqrt{2}, \quad N + F\sqrt{2} = \alpha^K,$$

where K tends to infinity through odd primes. Then N is odd, $N^2 - 2F^2 = -1$, and $N \asymp \lambda^{K/2}$. The moduli need not be prime. Use the phase-normalized cat quantization U_0 from the finite Floquet construction, so that $U_0 W(v) U_0^* = W(Tv)$ and $U_0^K = I$. Let

$$W(a,b)f(x) = e_N(a(x - b/2))f(x - b), \qquad B_v = W(v) + W(-v).$$

All symmetrized observables and U_0 preserve odd parity. Write $d = (N-1)/2$ and $\tau_-(M) = d^{-1} \operatorname{Tr}_- M$.

The perturbation is now a different, fixed Fourier observable:

$$A = B_{(1,0)}, \quad D = U_0 A U_0^* = B_{(3,4)}, \quad H = A - D, \quad V_\varepsilon = e^{i\varepsilon H} U_0. \tag{1}$$

It is a coboundary: $\sum_{j=0}^{K-1} U_0^j H U_0^{-j} = 0$ exactly. The real classical Fourier symbol of H is independent of N. The products of the two coordinates of (1,0), (3,4), (2,4), and (4,4) are even, so the centered finite-Weyl convention introduces no N-dependent sign for these fixed Fourier modes. As throughout the project, the quantum exponent epsilon corresponds to a classical action amplitude proportional to epsilon/N, not epsilon itself.

Put

$$s_N = \sin(4\pi/N) > 0, \quad \zeta = \varepsilon^2 s_N, \quad t_K = 2\zeta\sqrt{K}.$$

Theorem 1 (second-order critical law). If $t_K \to t \in [0, \infty)$, then for every fixed integer m,

$$\tau_-(V_{\varepsilon_K}^{mK}) \longrightarrow e^{-m^2 t^2/2}. \tag{2}$$

Thus the empirical spectrum of $V_{\varepsilon_K}^K$ tends to the wrapped normal law of variance t^2, or to the point mass at 1 when t=0. The critical quantum scale is

$$|\varepsilon_K| = \left(\frac{t}{2 s_N \sqrt{K}}\right)^{1/2} \sim \sqrt{\frac{t}{8\pi}}\, \sqrt{N} K^{-1/4} \quad (t > 0). \tag{3}$$

This can grow without bound, even though the classical amplitude tends to zero. The sign of epsilon does not change the limit in (2).

Theorem 2 (negative microscopic consequences). For the original propagator in (1), on the whole odd space and with global mean-spacing normalization, the following hold.

If $|\varepsilon_K|K^{1/4}/\sqrt{N} \to 0$, the empirical nearest-neighbour spacing measures tend weakly to δ_0, and the normalized number of ordered distinct pairs at distance at most x mean spacings tends to infinity for every fixed x>0.

For any subsequence on which $|\varepsilon_K|K^{1/4}/\sqrt{N}$ is bounded, the pair measures cannot converge to a bounded density R(s) with $R(s) \to 1$ as $|s| \to \infty$. In particular, escape from this obstruction requires

$$|\varepsilon_K|K^{1/4}/\sqrt{N} \longrightarrow \infty. \tag{4}$$

Condition (4) is only necessary. No positive universality theorem beyond it is proved. These statements concern global unfolding of the specified odd spectrum, not arbitrary clusterwise unfolding or an assertion that all arithmetic symmetries are resolved. They do not refute fixed-system BGS.

1. A second-order loop estimate valid at large quantum exponents

Lemma 3. For finite Hermitian A,D and any real epsilon,

$$\left\| e^{-i\varepsilon A} e^{i\varepsilon(A-D)} e^{i\varepsilon D} - e^{-\varepsilon^2[A,D]/2} \right\|_{\mathrm{op}} \le |\varepsilon|^3 \left(\frac{1}{2}\|[A,[A,D]]\|_{\mathrm{op}} + \frac{1}{6}\|[D,[A,D]]\|_{\mathrm{op}} \right). \tag{5}$$

There is no hypothesis $|\varepsilon|\|A\| \ll 1$.

Proof. Write $Q = [A, D]$ and $L(u) = e^{-iuA}e^{iu(A-D)}e^{iuD}$. Direct differentiation gives

$$L'(u)L(u)^* = -\int_0^u \mathrm{Ad}_{e^{-iuA}e^{iv(A-D)}}\, Q\, dv.$$

The logarithmic derivative of $E(u) = e^{-u^2Q/2}$ is -uQ. By differentiating unitary conjugation and integrating,

$$\|\,\mathrm{Ad}_{e^{-iuA}e^{iv(A-D)}}\, Q - Q\| \le u\|[A,Q]\| + v\|[A-D,Q]\| \quad (0 \le v \le u).$$

Duhamel comparison of the two unitary evolutions, integrated over $0 \le v \le u \le \varepsilon$, is at most $\varepsilon^3(\|[A,Q]\|/3 + \|[A-D,Q]\|/6)$. The triangle inequality proves (5) for nonnegative epsilon. Replace A,D by -A,-D for the other sign. This argument is a norm estimate, not a truncation of a formal BCH series. End of proof.

Let u=(2,4), z=(4,4) and C=B_u-B_z. Since $W(v)W(w) = e_N(\omega(v,w)/2)W(v+w)$ and $\omega((1,0),(3,4)) = 4$, exact multiplication yields

$$[A, D] = -2is_N C. \tag{6}$$

Each double commutator in (5) has at most eight Weyl terms of coefficient modulus $4s_N^2$. Therefore both norms are at most $32s_N^2$, and

$$\left\| e^{-i\varepsilon A} V_\varepsilon e^{i\varepsilon A} - e^{i\zeta C} U_0 \right\|_{\mathrm{op}} \le R_N(\varepsilon) := \frac{64}{3}|\varepsilon|^3 s_N^2. \tag{7}$$

The same bound holds in normalized odd Hilbert–Schmidt norm. At the scale (3), $KR_N = O_t(K^{1/4}/\sqrt{N})$; this tends to zero. The small quantity is controlled by nested commutators, not by the norm of epsilon A.

2. A two-species arithmetic lemma: collisions must not be discarded

Identify an integer vector v=(a,b) with $\psi(v) = b + \sqrt{2}a$. Then

$$\psi(Tv) = \lambda\psi(v), \quad \psi(u) = 4 + 2\sqrt{2}, \quad \psi(z) = \sqrt{2}\psi(u), \quad \mathrm{Norm}(\psi(u)) = 8.$$

Since N is odd, multiplication by $\psi(u)$ is invertible in $(\mathbb{Z}/N\mathbb{Z})[\sqrt{2}]$. The two species of time translates are therefore faithfully represented by λ^j and $\sqrt{2}\lambda^j$. Also $T^K = I$ modulo N, as follows from $Q_{2K} = 2N^2 + 1$ and $P_{2K} = 2NF$. Rotating all labels modulo K preserves a vector relation.

Lemma 4 (lifting). A signed m-atom modular zero relation in these two species, after cancellation of identical opposite atoms and rotation across a largest cyclic gap, has the form

$$P(\lambda) = \sum_{\nu} \epsilon_\nu b_\nu \lambda^{j_\nu}, \quad b_\nu \in \{1, \sqrt{2}\}, \quad 0 \le j_\nu \le K - \lceil K/m \rceil.$$

For sufficiently large K at fixed m this expression is zero in $\mathbb{Z}[\sqrt{2}]$, not just modulo N. For m=4, K>=11 suffices.

Proof. There are at most m occupied time labels, hence a cyclic gap of at least ceil(K/m). Modular vanishing and the invertibility just proved imply $P(\lambda) \in N\mathbb{Z}[\sqrt{2}]$. Its nonzero algebraic norm would thus have absolute value at least N^2, whereas the two real embeddings give

$$|\,\mathrm{Norm}\, P(\lambda)| \le 2m^2\lambda^{K-\lceil K/m \rceil}.$$

Since $N^2 = \lambda^K(1-\lambda^{-K})^2/4$, the latter is smaller than N^2 for large K at fixed m. For m=4 it suffices that $\lambda^{\lceil K/4 \rceil}(1-\lambda^{-K})^2 > 128$. At K>=11 this follows from $\lambda > 11/2$, ceil(K/4)>=3, and $1 - \lambda^{-K} > 9/10$, since $(11/2)^3(9/10)^2 > 128$. End of proof.

A four-atom relation in characteristic zero need **not** balance within individual species. Apart from cancellation, the sole type is

$$z_{j+1} - u_{j+1} = z_j + u_j, \qquad v_j := T^j v, \tag{8}$$

up to translation and multiplication by -1. The identity is already true over the integers, since $T(z-u) = z + u$.

Here is a direct completeness proof. Look at the maximal exponent E after cancellation. With one atom at E, its coefficient has modulus at least 1, whereas the three lower atoms sum to at most $3\sqrt{2}/\lambda < 1$ times λ^E. With three atoms at E, their nonzero coefficient has modulus at least $2 - \sqrt{2} > \sqrt{2}/\lambda$, which the remaining lower atom cannot cancel. With four atoms at E, linear independence of 1 and sqrt(2) forces balance. With two atoms at E, the only possible nonzero coefficient small enough is $\pm(\sqrt{2}-1)$. The lower atoms must both be at E-1: otherwise at that scale their modulus is at most $\sqrt{2}(1+\lambda^{-1}) < 1 + \sqrt{2}$. At E-1 they must be one of each species with equal signs because $\lambda(\sqrt{2}-1) = 1 + \sqrt{2}$. This is exactly (8). Relations with fewer occurrences reduce to the same cases; there is no nontrivial relation with at most three atoms.

Corollary 5 (multiplicity two). For K>=11, form all signed unordered pairs with distinct time labels:

$$\epsilon v_t + \delta w_s, \quad 0 \le t < s < K, \quad v, w \in \{u, z\}, \quad \epsilon, \delta \in \{-1, 1\}.$$

There are $8K(K-1)$ such pairs. Their vector keys are nonzero, have multiplicity at most two, and have precisely

$$4K \text{ double keys}, \quad 8K(K-2) \text{ single keys}, \quad 8K^2 - 12K \text{ distinct keys}. \tag{9}$$

The double classes are the two identities

$$z_{j+1} - z_j = u_{j+1} + u_j, \qquad z_{j+1} - u_j = u_{j+1} + z_j,$$

with their negatives, for j modulo K. The third partition of (8) uses equal-time pairs and is not in the domain. Each displayed signed pair belongs to only one class: its species, the two adjacent cyclic labels, and its signs determine the other pair. Thus there are no triple classes or overlapping equivalence chains. Lemma 4 ensures that no additional modular relation is missed. K=3 is a useful negative control: it has six triple keys, and the range restriction must not be removed.

3. Weighted phase control and a finite normal form for C

Write $C_j = U_0^j C U_0^{-j}$, $\bar{C} = K^{-1}\sum_j C_j$, $G_C = \sum_j c_j C_j$, where $c_j = (K-1-2j)/(2K)$. Then

$$G_C - U_0 G_C U_0^* = C - \bar{C}, \qquad [\bar{C}, U_0] = 0. \tag{10}$$

Define a phase majorant, including both species,

$$a_j = \max_{v,w\in\{u,z\}} |e_N(\omega(v, T^j w)) - 1|.$$

One has $a_{K-j} = a_j$. In particular, no independence assumption about the several phase factors is being made.

For a centered representative delta of j modulo K,

$$\omega(v, T^\delta w) = (2v_1 w_1 - v_2 w_2) P_{2\delta} + \det(v, w) Q_{2\delta}.$$

For the four choices of species these are $-8P_{2\delta}$, $16P_{2\delta}$, and $\pm 8Q_{2\delta}$, respectively. Their absolute values are at most $8\lambda^{|\delta|}$. If $|\delta| = (K-1)/2 - \ell > 0$, this gives

$$a_j \le \frac{32\pi}{\alpha(1-\lambda^{-K})}\lambda^{-\ell}.$$

At zero, $a_0 \le 16\pi/N$. Summing both signs, including their weights ℓ/K and $(\ell+1)/K$, proves the explicit estimates

$$\begin{aligned} L_K := \sum_j a_j &\le \frac{32\pi}{1-\lambda^{-K}} + \frac{16\pi}{N} < 128, \\ K\sum_j |c_j| a_j &\le \frac{16\sqrt{2}\pi}{1-\lambda^{-K}} + \frac{8\pi K}{N} < 128. \end{aligned} \tag{11}$$

For K>=3, use $\lambda^{-K} < 1/10$, $N \ge 2K+1$, $\pi < 22/7$, and $\sqrt{2} < 3/2$ to check the last inequalities. These constants are deliberately conservative.

The commutator $[C_t, C_s]$ has sixteen Weyl terms, each of modulus at most a_{s-t}. Corollary 5 says that each key in the sum over t<s has at most two preimages. Weyl matrices are orthonormal in the *full* normalized Hilbert–Schmidt inner product, also over composite odd N. Cauchy–Schwarz inside each key class therefore gives

$$\begin{aligned} \|[G_C, \bar{C}]\|_{2,N}^2 &\le \frac{32}{K^4}\sum_{j=1}^{K-1}(K-j)j^2 a_j^2 \\ &= \frac{16}{K^3}\sum_{j=1}^{K-1} j(K-j)a_j^2 \le \frac{1024}{K}. \end{aligned} \tag{12}$$

The equality between the two sums uses $a_{K-j} = a_j$; the final bound uses $a_j^2 \le 2a_j$ and (11). Norm restriction to the odd subspace obeys $\|M\|_{2,-} \le \sqrt{2N/(N-1)}\|M\|_{2,N}$ by positivity. We have not assumed orthogonality of restricted Weyl matrices.

The triangle inequality in operator norm also gives

$$\|[G_C, C]\|_{\mathrm{op}} \le 2048/K, \qquad \|[C, \bar{C}]\|_{\mathrm{op}} \le 2048/K. \tag{13}$$

The usual two-exponential Duhamel inequality is $\|e^{ixA}e^{ixD} - e^{ix(A+D)}\| \le x^2\|[A,D]\|/2$ in any unitarily invariant norm. Applying it first to -G_C,C, then to C-G_C and $U_0 G_C U_0^*$, using (10)–(13), proves

$$\left\| e^{-i\zeta G_C}(e^{i\zeta C}U_0)e^{i\zeta G_C} - e^{i\zeta \bar{C}}U_0 \right\|_{2,-} \le \frac{1000\zeta^2}{\sqrt{K}} \quad (K \ge 11). \tag{14}$$

Indeed the less rounded bound is $\zeta^2(2048/K + 16\sqrt{2N/(N-1)}/\sqrt{K})$, which is smaller than (14) already using $K \ge 11$ and $N \ge 7$. This is another estimate valid for every real zeta, not a formal expansion requiring a small generator.

Combining the two gauges and telescoping unitary powers gives

$$\left|\tau_-(V_\varepsilon^{mK}) - \tau_-(e^{im\zeta K\bar{C}})\right| \le |m|KR_N(\varepsilon) + 1000|m|\zeta^2\sqrt{K}. \tag{15}$$

4. The two-species Gaussian limit

Lemma 6. Put $X_K = (\sqrt{K}/2)\bar{C}$. For every fixed r>=1,

$$\tau_-(X_K^{2r}) = (2r-1)!! + O_r(K^{-1}) + O_r(K^r/N). \tag{16}$$

For every fixed odd m, eventually $|\tau_-(X_K^m)| \le (4K)^{m/2}/(N-1)$. Consequently its empirical spectral measures converge weakly and in quadratic Wasserstein distance to standard normal.

Proof. There are 4K signed Weyl atoms, with coefficient +1 for species u and -1 for species z. For a word with central phase e_N(phi) and total index v, the exact odd trace is

$$e_N(\phi)\frac{N\mathbf{1}_{v=0} - 1}{N-1}. \tag{17}$$

This follows from $\operatorname{Tr} W(v) = N\mathbf{1}_{v=0}$ and $\operatorname{Tr}(RW(v)) = 1$. The noncentral contribution after normalization $(4K)^{-m/2}$ is at most $(4K)^{m/2}/(N-1)$.

Apply Lemma 4 to central words. The following bounded-cluster argument handles the unavoidable identities (8), rather than incorrectly excluding them. In a nonzero polynomial relation with at most m signed atoms and coefficients 1 or sqrt(2), a gap d with $\lambda^d > 2m^2$ separates independently vanishing components. To see this, write $P = P_{\mathrm{low}} + Z^{a+d}P_{\mathrm{high}}$ with degree of P_low at most a. If $P_{\mathrm{high}}(\lambda) \ne 0$, its nonzero integer norm and the bound $|P_{\mathrm{high}}(\lambda^{-1})^{\mathrm{conjugate}}| \le \sqrt{2}m$ give $|P_{\mathrm{high}}(\lambda)| \ge 1/(\sqrt{2}m)$. This contradicts cancellation by $|P_{\mathrm{low}}(\lambda)| \le \sqrt{2}m\lambda^a$. Iterating produces components with spans bounded solely in m.

Cancel opposite identical atoms first and restore these pairs after splitting. Every nonzero component then contains at least three occurrences. A two-atom relation cannot change species, because its two algebraic norms would be 1 and -2, and cannot change exponent, because lambda>1. A nonbalanced 2r-word thus has at most r-1 components. There are only finitely many partitions of its positions, species, signs and bounded offset patterns at fixed r. Each component has K possible cyclic translations, so there are $O_r(K^{r-1})$ such words. Choosing the largest-gap cut introduces no additional K factor: this count is over the actual translated components. Their normalized contribution is $O_r(1/K)$.

In a balanced word, pair identical species-time labels with opposite signs. For each pairing there are K^r time choices, 2^r species choices and 2^r sign choices. Their product is precisely the normalization $(4K)^r$. Repeated pair labels cause at most $O_r(K^{r-1})$ overcount. Signs from species z occur twice in each pair and hence multiply to +1. The Weyl phase is a product of crossing factors or their conjugates. For fixed species, each crossing time difference is uniform on

the cyclic group. By (11) and $|\prod u_j - 1| \le \sum |u_j - 1|$, its average error is at most 128 times the number of crossings divided by K. No independence of the crossing factors is assumed. Summing over the $(2r-1)!!$ pairings gives (16).

For odd m, reduce a lifted relation in $\mathbb{Z}[\sqrt{2}]$ modulo 2. Here $\lambda = 3 + 2\sqrt{2}$ reduces to 1. The coefficients of both 1 and sqrt(2) in the relation must be even. Both species counts would therefore be even, contradicting odd m. Thus no odd central word exists after lifting.

Bounded second moments yield tightness, higher even moments yield uniform integrability, and the Gaussian moment sequence is determinate, since its Carleman sum diverges. Therefore all weak subsequential limits are normal. Convergence of the second moments gives the claimed Wasserstein upgrade. End of proof.

Since $\zeta K\bar{C} = t_K X_K$, (15) and Lemma 6 prove Theorem 1. More explicitly, the finite comparison error is bounded by

$$|m| \left(K R_N(\varepsilon) + \frac{250 t_K^2}{\sqrt{K}} \right). \tag{18}$$

For bounded t_K, $KR_N = O(K^{1/4}/\sqrt{N})$ follows directly from $\varepsilon^2 = t_K/(2s_N\sqrt{K})$ and $s_N \le 4\pi/N$. The remaining Gaussian characteristic-function error is o(1); no universal $K^{-1/2}$ rate is asserted for that separate step.

5. Microscopic consequences without conflating unfolding conventions

For completeness we give the two spectral implications used in Theorem 2. First, the 4K Weyl atoms in C_j are distinct, so $\mathrm{Tr}(\bar{C}^2)/N = 4/K$. Distinctness follows within species from the unit conditions on $1 \pm h^j$ in the Pell construction. Between species, norm 1 versus -2 would require N to divide 3, which is impossible for K>=3. Consequently

$$\tau_-(\bar{C}^2) \le \frac{8N}{K(N-1)}.$$

The squared-overlap matrix between orthonormal eigenbases of two unitaries is doubly stochastic. Decomposing it into permutations yields an eigenvalue matching whose mean squared chord cost is bounded by normalized squared Hilbert–Schmidt distance. Circle distance in $\mathbb{R}/\mathbb{Z}$ is at most one quarter of chord distance. Using this matching, (7),(14), and $[U_0, \bar{C}] = 0$, the RMS distance a of the spectrum of V_epsilon from the K-point lattice satisfies

$$a \le \frac{R_N(\varepsilon)}{4} + \frac{250\zeta^2}{\sqrt{K}} + \frac{|\zeta|}{2\pi}\sqrt{\frac{8N}{K(N-1)}}. \tag{19}$$

When t_K tends to zero, Ka tends to zero, and K/d tends to zero.

For an arbitrary d-point circle multiset with this RMS lattice distance, placing arcs of half-width w<1/(2K) around the lattice and using Markov's inequality gives, for its unfolded cyclic gaps S_l,

$$d^{-1}\#\{l : S_l > x\} \le 2a^2/w^2 + K/d + 2Kw/x. \tag{20}$$

There are at most da^2/w^2 bad points; at most twice as many gaps touch one, at most K other gaps leave an arc, and the within-arc gaps have total length at most 2Kw. This proves (20). Partitioning the arcs into cells of length at most x/d and applying Cauchy–Schwarz to their occupancies also gives the ordered close-pair bound

$$R_2(x) \ge \frac{\max(0, 1 - a^2/w^2)^2}{K/d + 2Kw/x} - 1. \tag{21}$$

Take $w = (Ka)^{2/3}/K$ for (20), and w=2a for (21). These prove subcritical spacing collapse and divergent close-pair counts. The a=0 case follows by counting at most K nonzero gaps and using $R_2(x) \geq d/K - 1$. The average spacing remains exactly one: weak convergence to delta_0 does not imply convergence of first moments, which are carried by rare long gaps.

For bounded nonzero critical parameter a different argument is necessary. Let $\phi_L(s) = L^{-1}(1 - |s|/L)_+$, a compactly supported nonnegative function with integral one and Fourier transform $\mathrm{sinc}^2(L\xi)$. For the eigenangles theta_l in $\mathbb{R}/\mathbb{Z}$, define

$$C_d(\phi_L) = \frac{1}{d} \sum_{l \neq j} \sum_{q \in \mathbb{Z}} \phi_L(d(\theta_l - \theta_j + q)).$$

Poisson summation, including the zero Fourier mode and the removed diagonal, proves, for d>L,

$$C_d(\phi_L) \geq 1 - \frac{1}{L} + 2 \sum_{m=1}^{M} \mathrm{sinc}^2(LmK/d) |\tau_-(V_\varepsilon^{mK})|^2. \tag{22}$$

All omitted Fourier coefficients are nonnegative. Take K to infinity at fixed M, use (2), then let M grow. For t>0 this gives

$$\liminf C_d(\phi_L) \geq \Theta(t) - 1/L, \qquad \Theta(t) = \sum_{m \in \mathbb{Z}} e^{-m^2 t^2} > 1. \tag{23}$$

For t=0 the lower bound is infinite. If the pair measures converged on compact tests to a bounded density R(s) tending to 1 at infinity, then $\int \phi_L R \to 1$ as L grows. This contradicts (23). A bounded subsequence of $|\varepsilon| K^{1/4}/\sqrt{N}$ has a further subsequence with finite limiting t_K, so (4) follows. This is a necessary-condition argument, not an exchange of a growing test support with the semiclassical limit.

An equivalent moving-window consequence is

$$\liminf \mathrm{Var}_\theta[\#\{l : \theta_l \in [\theta, \theta + L/d)\}] \geq L^2(\Theta(t) - 1).$$

Indeed the finite variance is $L^2(C_d(\phi_L) - 1) + L$. The underlying positive Fourier test was already established in the earlier Floquet note; it is not claimed as a new result of this pass.

6. Scientific boundary and verification contract

The scale (3) is parametrically larger than the first-order cosine scale $\varepsilon \asymp K^{-1/2}$, but the observable is different. This is not an improvement obtained while silently changing a fixed hypothesis: the role of the coboundary is explicit in (1). Both classical perturbation amplitudes vanish. No fixed nonzero classical perturbation, positive BGS universality, or trace theorem at times of order d is proved.

The new ingredients relative to the maintained project are the uniform loop bound with growing quantum exponent, the multiplicity-two classification needed for the generated two-species observable, and their deterministic spectral combination. The residual-perturbation mechanism and its second-order commutator have direct prior literature. A full priority assessment remains unfinished; no prize-level or first-ever claim follows from these notes.

Verification uses `pell_coboundary_regression.py` in the verification directory and the independent `pell_coboundary_pairs.cpp` in the C++ directory. The Python pair verifier iterates original T-vectors and checks each double class, while C++ uses the quotient quadratic ring after dividing out 4+2sqrt(2). Floating-point dense checks test (5)–(7) and the complete propagator at K=3,5,7, explicitly outside the universal K>=11 collision range. They are separate from exact combinatorial checks and not an asymptotic proof. The finite Lean certificate has its own narrower scope; its successful compilation must be recorded rather than inferred from source code.

Numerical propagation and its separate error budget

The optional large-N experiment applies the actual coboundary kick rather than replacing it by the effective observable. Since the norm of H is at most 4, the Jacobi–Anger expansion gives

$$e^{i\varepsilon H} = J_0(z)I + 2\sum_{j\geq 1} i^j J_j(z) T_j(H/4), \qquad z = 4\varepsilon.$$

Here T_j is a Chebyshev polynomial. The standard Bessel series, as recorded in NIST DLMF 10.2.2, implies

$$|J_j(z)| \leq b_j := \frac{(|z|/2)^j}{j!} \exp\left(\frac{z^2}{4(j+1)}\right).$$

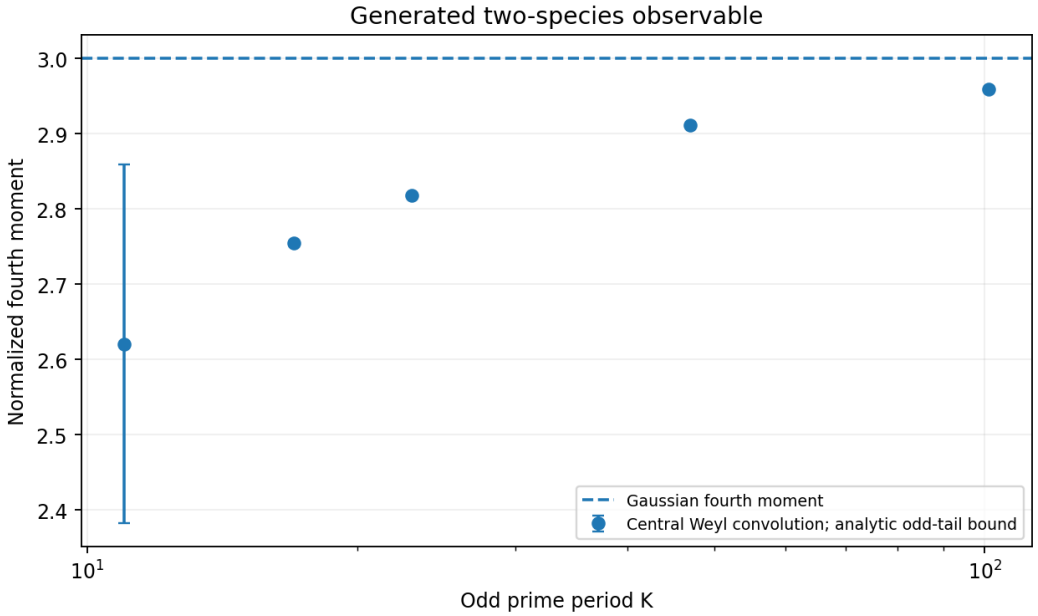
For $M + 2 > |z|/2$, the discarded operator tail is consequently bounded by

$$2\sum_{j>M} |J_j(z)| \leq \frac{2b_{M+1}}{1 - |z|/(2(M+2))}.$$

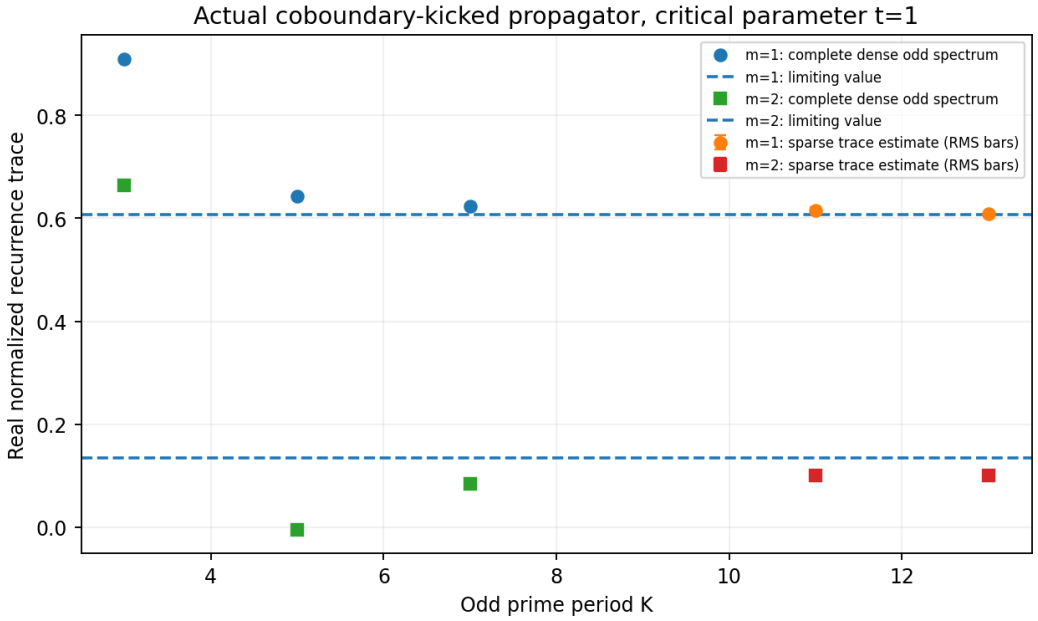
The code chooses the degree using this bound and checks norm preservation. This is a truncation estimate, not interval certification of Bessel-function rounding or of subsequent matrix arithmetic. A separate exponential-action implementation agreed on the recorded K=11 controls to about 1e-13.

For R independent probes with fourth-root-of-unity coefficients in an orthonormal odd basis, the normalized trace estimator has complex mean-square error at most $1/(Rd)$ for exact unitary applications. Indeed, only off-diagonal entries contribute to its variance and their squared sum is at most d. The displayed RMS bounds are stochastic, not confidence intervals. First and second recurrence modes, including finite-size bias, are both retained. No agreement with a finite simulation is used as a proof of the limiting theorem.

Reproducible diagnostics



Central fourth moments are evaluated by Weyl convolution, not matrix diagonalization. Bars bound the noncentral odd trace analytically; floating-point evaluation of the center is not interval-certified.



Both recurrence modes are retained, including the visible finite-size bias of the second mode. Sampled bars are complex RMS bounds over independent random probes, not confidence intervals or deterministic error bars.